Geometric Truncation of Low-Multipole CMB Power and Null B-Mode Prediction from a QCD-Scale Euclidean Instanton Bounce

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Repository: https://github.com/infosave2007/vmf

Abstract

The persistent lack of power at low multipoles ($\ell \leq 10$) in the Cosmic Microwave Background (CMB) temperature anisotropy spectrum is traditionally dismissed as "cosmic variance" within the standard ΛCDM cosmological model. We propose that this anomaly is not a statistical fluctuation, but a strict physical truncation resulting from the universe emerging from a finite Euclidean instanton bounce at the Quantum Chromodynamics (QCD) density scale ($\rho_c \sim 10^4$ MeV/fm^3), rather than from a point-like Big Bang singularity. Using the Null-Vector Gravity (NVG) framework, we analytically derive the radius of this Genesis instanton ($r_c \approx 1.13$ km) and the exact required number of e-folds ($N_e = \ln(R_{H0}/r_c) \approx 53.2$), bypassing the arbitrary initial conditions and scalar fields of standard inflation. The maximum comoving wavelength of primordial perturbations is strictly bounded by the physical size of the instanton, resulting in an exact mathematical suppression of the quadrupole ($\ell = 2$) and octupole ($\ell = 3$). Furthermore, because the bounce occurs at an energy scale $\sim 10^{77}$ times lower than the Planck scale, primordial tensor perturbations are not excited, resulting in a strict null prediction for the tensor-to-scalar ratio ($r \approx 0$). This provides a definitive, falsifiable target for next-generation B-mode observatories such as LiteBIRD.

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1. Introduction

Precision observations of the Cosmic Microwave Background (CMB) by the WMAP and Planck satellites have firmly established the ΛCDM model as the standard paradigm of cosmology. However, these same observations have repeatedly highlighted persistent anomalies at the largest angular scales (low multipoles, $\ell \leq 10$). Specifically, the measured amplitude of the quadrupole ($\ell = 2$) and octupole ($\ell = 3$) is substantially lower than the predictions of a nearly scale-invariant primordial power spectrum generated by standard slow-roll inflation.

In the absence of a theoretical mechanism to suppress large-scale power, this discrepancy is typically attributed to "cosmic variance"—the inherent statistical uncertainty in observing only one realization of the universe at the largest scales. However, if this suppression is a physical feature rather than a statistical fluke, it points toward new physics prior to or during the earliest moments of expansion.

Simultaneously, the inflationary paradigm, while successful at solving the horizon and flatness problems, requires the ad-hoc introduction of an inflaton scalar field and suffers from the "initial conditions" problem. Inflation typically operates at energy scales near the Grand Unified Theory (GUT) scale ($V \sim 10^{16}$ GeV), inherently predicting a background of primordial gravitational waves (B-modes) parametrized by the tensor-to-scalar ratio r. Despite intense observational efforts, r remains undetected, with current upper bounds $r < 0.036$.

In this letter, we apply the Null-Vector Gravity (NVG) cyclic framework to demonstrate that a macroscopic, semi-classical bounce at the QCD phase transition scale simultaneously solves the horizon problem without inflation, mathematically explains the low-ℓ CMB anomaly as a geometric truncation, and yields a rigid null prediction for r.

2. The QCD-Scale Euclidean Instanton Bounce

In the NVG framework, the structural mass of fermions is treated as a dynamic coupling to a macroscopic vacuum condensate, anchored by the empirical non-perturbative QCD mass gap ($M_{\Omega,0} = 859$ MeV). During the contraction phase of a cyclic universe, as the global energy density approaches the QCD phase transition threshold ($\rho_c \approx 7.09 \times 10^4$ MeV/fm^3), the vacuum mass fraction melts entirely. The strong energy condition ($\varepsilon + 3P \geq 0$) is violated globally, and the universe transitions into a locally de Sitter-like state.

This phase transition prevents a classical singularity. Instead, the universe undergoes a "Genesis" bounce via a Euclidean instanton. Because the density is bounded by ρ_c, the minimum physical radius of the universe at the moment of the bounce is strictly finite and non-zero. Our derivation yields the instanton radius:

$$r_c \approx 1.13 \text{ km}$$

This macroscopic initial size ($\sim 10^{38}$ times larger than the Planck length) ensures that the bounce occurs entirely within the domain of semi-classical gravity, avoiding the need for a full theory of quantum gravity.

3. Analytic Derivation of e-folds (N_e)

Standard inflation requires an arbitrary parameter of $N_e \sim 50 - 60$ e-folds of exponential expansion to solve the horizon problem. In the NVG model, the number of e-folds is not a free parameter but a geometric necessity derived from the ratio of the current Hubble horizon R_{H0} to the initial instanton radius r_c.

Assuming standard radiation and matter domination expansion histories following the bounce, the topological expansion required to reach the current observable universe is exactly analytically constrained:

$$N_e = \ln\left(\frac{R_{H0}}{r_c}\right) \approx 53.2$$

This derivation provides the exact expansion factor required to generate a flat, homogeneous observable universe today, originating purely from the physical constraints of the QCD vacuum condensate, without the need to postulate a transient inflaton field.

4. Geometric Truncation of Low Multipoles

The finite size of the initial instanton has profound consequences for the primordial power spectrum. In standard inflation, quantum fluctuations are stretched from sub-Planckian scales to super-horizon scales, theoretically allowing perturbations of infinite wavelength.

In the NVG Genesis model, the maximum physical size of any comoving perturbation is strictly bounded by the circumference of the initial instanton, $2\pi r_c$. A perturbation with a wavelength larger than the universe itself at the time of its generation cannot exist. This imposes a hard physical cutoff on the primordial power spectrum $P(k)$ at $k_{min} \sim 1/r_c$.

When mapped to the CMB angular power spectrum C_ℓ, this wavelength truncation manifests as a sharp suppression of power at the largest angular scales. Our numerical integration of this geometric cutoff yields: * **Quadrupole ($\ell = 2$) Power:** Suppressed to $\sim 28\%$ of the standard ΛCDM expectation. * **Octupole ($\ell = 3$) Power:** Suppressed to $\sim 52\%$ of the standard ΛCDM expectation.

These values align closely with the observed anomalies in the Planck and WMAP datasets, providing a deterministic, physical explanation for a phenomenon previously dismissed as cosmic variance.

5. Null Prediction for Primordial B-Modes

The amplitude of primordial tensor perturbations (gravitational waves) generated during the early universe is directly proportional to the energy scale of the expansion/bounce mechanism, V. Standard inflation, operating at $V^{1/4} \sim 10^{16}$ GeV, predicts a tensor-to-scalar ratio $r \geq 10^{-3}$.

The NVG bounce, however, is driven by the melting of the QCD vacuum mass fraction at an energy scale corresponding to $T_b \approx 432$ MeV. The ratio of the NVG bounce energy density to the standard GUT inflationary scale is:

$$\frac{\rho_{NVG}}{\rho_{GUT}} \sim 10^{-77}$$

Because the bounce occurs at such a profoundly low energy scale, the excitation of primordial tensor modes is mathematically negligible. The model yields a strict, falsifiable null prediction for the tensor-to-scalar ratio:

$$r \approx 0 \quad \text{(bounded practically at } < 10^{-68}\text{)}$$

6. Conclusion

The NVG cyclic framework offers a paradigm shift in our understanding of the early universe. By anchoring the cosmological bounce to the empirically verified non-perturbative QCD mass gap rather than the speculative Planck scale, we analytically derive the required expansion e-folds ($N_e = 53.2$) without invoking scalar fields.

Most importantly, the macroscopic finite size of the Genesis instanton ($r_c = 1.13$ km) imposes a rigid geometric cutoff on large-scale perturbations, naturally explaining the CMB low-ℓ anomalies. Finally, the low energy scale of the bounce guarantees the absolute absence of primordial B-modes. We offer $r \approx 0$ as a definitive, zero-parameter theoretical target for the upcoming LiteBIRD mission. Detection of $r \geq 10^{-3}$ would definitively falsify this framework.

Data Availability

Full analytic derivations and Python computational suites modeling the Genesis instanton and CMB power suppression are publicly available at the NVG repository: [https://github.com/infosave2007/vmf] and archived on Zenodo [https://zenodo.org/records/20214457].

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